

Vectors and Coordinates #3

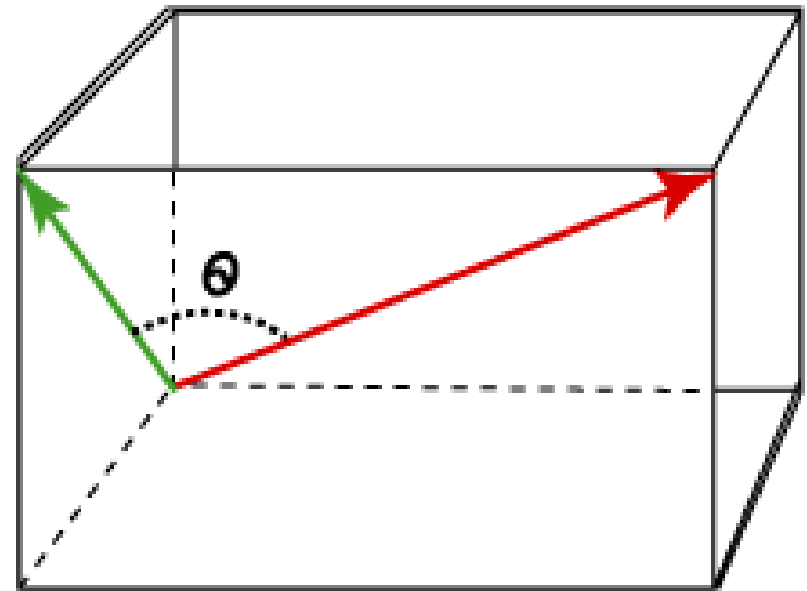


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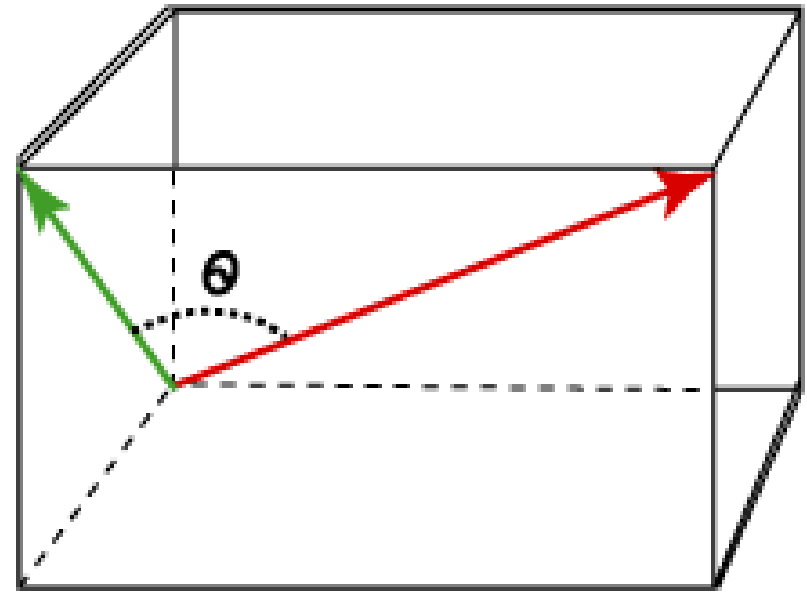
Angle Between Vectors

- If we want to work out the angle between two vectors, we need to use our dot product alongside our magnitude calculations
- The angle tells us how closely two vectors are aligned.
- This can help us consider design decisions and how forces react to each other



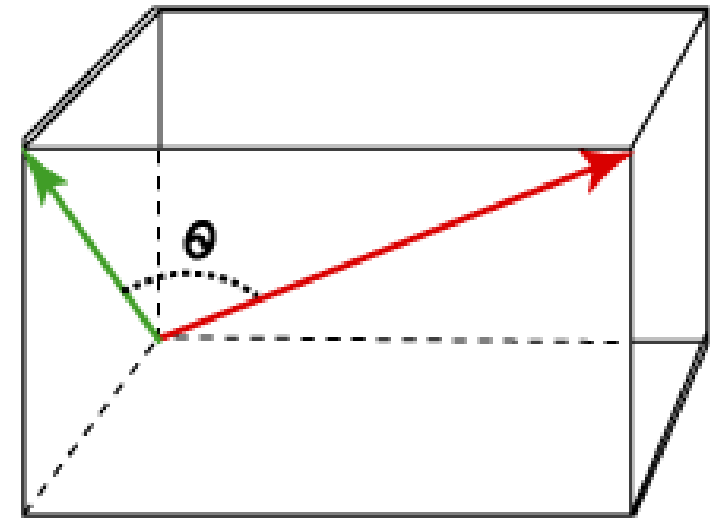
Angle Between Vectors

- To work out the angle between two vectors we do:
- $\theta = \cos^{-1} \left(\frac{a \cdot b}{|a||b|} \right)$
- The top requires us to work out the dot product whilst the bottom requires us to work out magnitudes



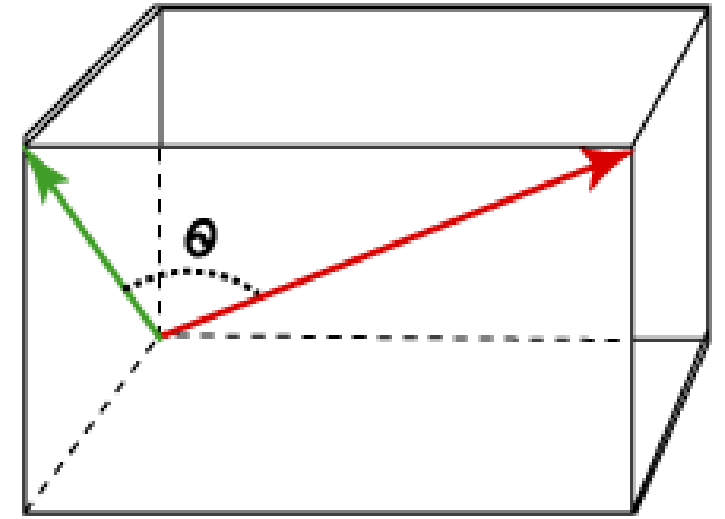
Example finding angle between (2D)

- Say we have two forces acting on a component $\mathbf{F1} = (3, 4)N$, $\mathbf{F2} = (5, 0)N$
- First, we calculate dot product: $F_1 \cdot F_2 = (3 * 5) + (4 * 0) = 15$
- Second, we calculate the magnitudes of both vectors:
 - $|F_1| = \sqrt{3^2 + 4^2} = 5$
 - $|F_2| = \sqrt{5^2 + 0^2} = 5$
- Finally, we sub into the equation:
 - $\theta = \cos^{-1} \left(\frac{a \cdot b}{|a||b|} \right) = \cos^{-1} \left(\frac{15}{5 \cdot 5} \right) = \cos^{-1} \left(\frac{15}{25} \right) \approx 53.1^\circ$



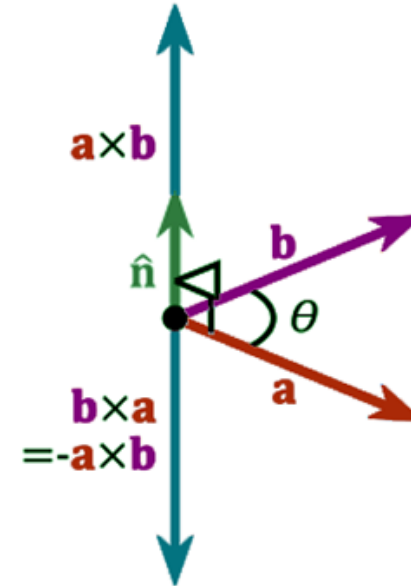
Example finding angle between (3D)

- Two force vectors act on a joint $\mathbf{A} = (2, -1, 3)$, $\mathbf{B} = (4, 0, -2)$
- First, we calculate dot product: $\mathbf{A} \cdot \mathbf{B} = (2 * 4) + (-1 * 0) + (3 * -2) = 2$
- Second, we calculate the magnitudes of both vectors:
 - $|\mathbf{F}_1| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{14}$
 - $|\mathbf{F}_2| = \sqrt{4^2 + 0^2 + (-2)^2} = \sqrt{20}$
- Finally, we sub into the equation:
 - $\theta = \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \right) = \cos^{-1} \left(\frac{2}{\sqrt{14} * \sqrt{20}} \right) = \cos^{-1} \left(\frac{2}{\sqrt{280}} \right) \approx 83.1^\circ$



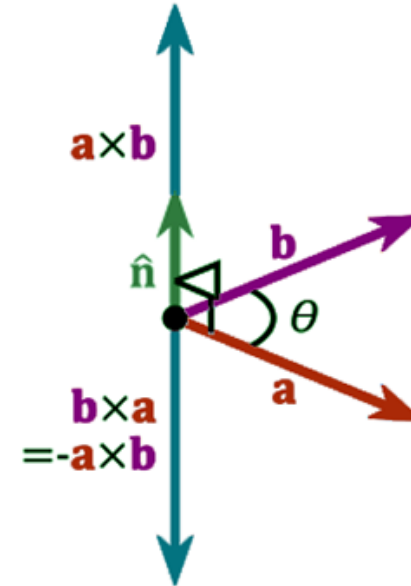
Cross Product

- The cross product combines two vectors and gives a new vector.
- $\mathbf{A} \times \mathbf{B}$
- The new vector is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.
- Why it matters:
 - Torque and moments
 - Finding a perpendicular direction
 - Rotational effects



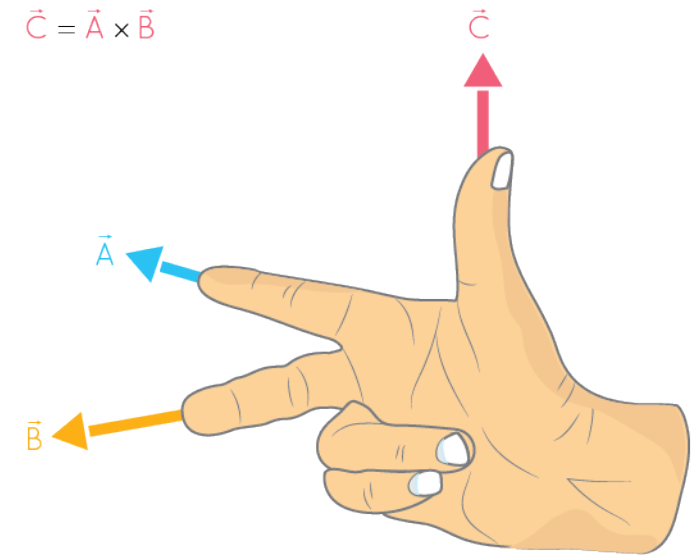
Cross Product

- The magnitude of the cross product can be calculated by:
- $|A \times B| = |A| |B| \sin \theta$
- Where **A** and **B** are the two vectors
- θ is the angle between them
- This means:
 - If vectors are parallel, cross product = 0
 - If vectors are perpendicular, cross product is maximum



Cross Product

- The direction of the cross product can be worked out using the right hand rule:
- Align your first and second finger with the vectors and your thumb will point in the direction the vector must apply



Cross Product

- We can calculate the overall dot product using this equation:

$$\begin{aligned} A &= (A_x, A_y, A_z) \\ B &= (B_x, B_y, B_z) \\ A \times B &= \begin{pmatrix} A_y B_z - A_z B_y \\ A_z B_x - A_x B_z \\ A_x B_y - A_y B_x \end{pmatrix} \end{aligned}$$

$$\vec{A} = [a, b, c] \quad \vec{B} = [d, e, f]$$

$\mathbf{i}$	$\mathbf{j}$	$\mathbf{k}$	$\mathbf{i}$	$\mathbf{j}$
a	b	c	a	b
d	e	f	d	e

- Or in unit vector:

$$\vec{A} \times \vec{B} = [(bf - ce), (cd - af), (ae - bd)]$$

$$A \times B = (A_y B_z - A_z B_y)\mathbf{i} + (A_z B_x - A_x B_z)\mathbf{j} + (A_x B_y - A_y B_x)\mathbf{k}$$

Example solving the cross product

Given:

$$A = (2, 1, 3)$$

$$B = (1, 4, 2)$$

$$\begin{aligned} A \times B &= \begin{pmatrix} (1)(2) - (3)(4) \\ (3)(1) - (2)(2) \\ (2)(4) - (1)(1) \end{pmatrix} \\ &= \begin{pmatrix} -10 \\ -1 \\ 7 \end{pmatrix} \end{aligned}$$

Your Turn

For each of these work out the magnitude and the dot product:

1. $A = (1, 2, 3), \quad B = (2, 1, 4)$

2. $A = (3, 1, 2), \quad B = (1, 4, 2)$

3. $A = (2, 3, 1), \quad B = (4, 1, 2)$

4. $A = (2, -1, 3), \quad B = (1, 4, -2)$

5. $A = (-3, 2, 1), \quad B = (4, -1, 5)$

6. $A = (3, -2, 4), \quad B = (-1, 5, 2)$

Extension

Two vectors are

$$A = (2, 1, 3), \quad B = (4, -2, 1)$$

- a) Calculate $A \times B$
- b) Calculate the magnitude of $A \times B$
- c) Explain what the direction of the resulting vector represents.